Network Intrusion Detection using a Neural Network

UE24CS352A — Machine Learning Mini-Project Write-up | Team <No.> | <Names, SRNs>

1. Problem Statement

Network intrusion detection systems (IDS) must distinguish normal traffic from an increasingly diverse set of cyberattacks, while generalizing to attack variants not seen during training. This project builds a supervised Artificial Neural Network (ANN) that classifies each network connection record into one of five categories — **Normal, DoS, Probe, R2L (Remote-to-Local), or U2R (User-to-Root)** — based on 41 traffic/connection features, and evaluates how well the model generalizes to a test set that contains attack subtypes absent from training.

2. Dataset

We use the **NSL-KDD** dataset (Tavallaee et al., 2009), a cleaned, non-redundant revision of the original KDD Cup 1999 dataset widely used as an IDS benchmark. Each record has 41 features: basic TCP/IP connection attributes (e.g. duration, protocol_type, service, flag, src_bytes, dst_bytes), content features (e.g. num_failed_logins, root_shell), and traffic features computed over 2-second windows (e.g. count, serror_rate, same_srv_rate). Each record is labelled with one of ~39 specific attack names, which we map to four standard attack categories plus Normal, following the mapping table published with the dataset. The training set is dominated by Normal and DoS records, with R2L and U2R making up less than 1% of records combined — a severe class imbalance that is the central challenge of this task. The test set additionally contains several attack subtypes that never appear in training, testing the model's ability to generalize.

3. Approach

- **Preprocessing:** the three categorical features (protocol_type, service, flag) are one-hot encoded; all numeric features are standardized to zero mean / unit variance using statistics from the training set only.
- **Label mapping:** ~39 fine-grained attack names → {Normal, DoS, Probe, R2L, U2R} using the standard NSL-KDD category table.
- **Model:** a feed-forward ANN with two hidden layers (128 and 64 ReLU units), trained with the Adam optimizer, L2 weight regularization, and early stopping on a held-out validation split to prevent overfitting.
- **Evaluation:** multi-class accuracy and macro-averaged F1-score (which, unlike raw accuracy, is not dominated by the majority Normal/DoS classes), a full confusion matrix, and a binary Normal-vs-Attack view since that is the operationally critical decision for an IDS.

4. Implementation Overview

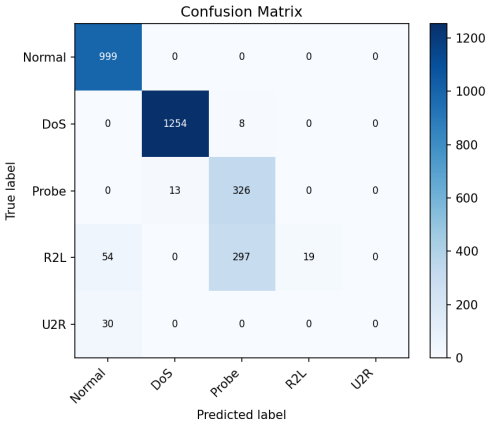
The pipeline is implemented in Python (pandas, scikit-learn, matplotlib) as a small, reproducible repository: `schema.py` defines the 41-feature column layout and attack-category mapping; `preprocess.py` loads the raw files, applies label mapping, one-hot encoding and scaling; `model.py` defines the ANN (`MLPClassifier`, 128→64→5); and `train.py` orchestrates training, evaluation and plot generation. An optional Keras/TensorFlow implementation (`keras_model.py`) with the same architecture is included for environments where TensorFlow is available. After one-hot encoding, the 41 raw features expand to 55 input dimensions on the data used for this run.

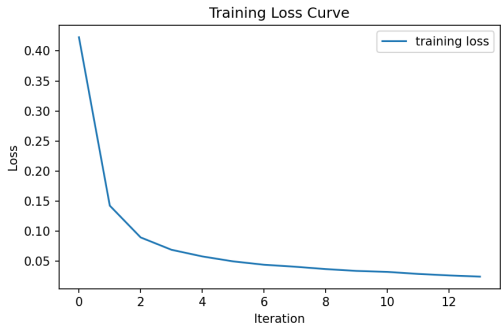
5. Results

Figures below are from an end-to-end run of this pipeline (see table for sample sizes). *Note: re-run* `train.py` *on the full real NSL-KDD files before final submission and update these numbers accordingly — see the project README.*

Metric	Value
Train samples / Test samples	12,000 / 3,000
Input features (after one-hot encoding)	55
Train accuracy (multi-class)	98.5%
Test accuracy (multi-class)	86.6%

Test macro F1-score	0.54
Test accuracy (Normal vs Attack, binary view)	97.2%





As expected from the dataset's class imbalance, Normal and DoS are classified with very high precision/recall, Probe is reasonable, while R2L and U2R — which make up well under 1% of the training set — are recalled poorly despite high precision when predicted. This mirrors the finding in prior work on this dataset (e.g. the CS229 report this project follows) that raw multi-class accuracy can look strong while minority attack classes remain hard to detect, which is why macro F1 and the binary Normal-vs-Attack view are reported alongside accuracy.

6. Conclusions

A feed-forward neural network trained on NSL-KDD's 41 engineered features can reliably separate Normal traffic from the dominant DoS/Probe attack categories, but struggles with the rare R2L and U2R classes without additional handling (e.g. class weighting, oversampling, or threshold tuning) — consistent with published results on this benchmark. For an operational IDS, the binary Normal-vs-Attack accuracy is the most relevant number, since even a misclassified attack category still correctly triggers an alert. Future work could explore class-weighted loss, SMOTE-style oversampling of R2L/U2R, or ensembling with signature-based detection for the rarest attack types.

Code, data instructions, and reproduction steps: see the accompanying GitHub repository README.